

Re: Application for the EngD position, Developing an Interactive Reverse Logistics Map for Circular Construction (CIRCOLOGIC), University of Twente

Dear Professor Hans Voordijk, Professor Berkhout, Professor Marc, and Members of the Selection Committee,

I am writing to apply for the EngD position on the interactive reverse logistics map for circular construction (CIRCOLOGIC). This is a topic I really care about. In construction we are already good at taking buildings apart, but a lot of good material still ends up as waste. The main problem is information. People often do not know which materials are available, in what condition, where they are, or when they will need them. So they choose new materials instead of reused ones. A map that shows this, a tool that calculates cost and CO2, and a simple way to compare options would make reuse a much easier choice. That is the kind of tool I would like to help build.

I first started thinking about this during my master's in environmental engineering. My thesis was about using Agile methods in construction projects. It showed me that good projects need clear communication and good information, not only good technology. After that, one question stayed with me: what happens to the materials when a building is taken down, and why is reuse still so hard? For me the answer came back to information and decisions. That is why I want to move from running projects to building the digital tools that help people make these decisions.

My work experience fits this project well. At Aug.e I worked on digital twins and learned how to bring data from buildings and sensors together, so people can use it to make decisions. At Hive CPQ I worked with large data sets and turned complicated rules into simple tools that customers could actually use, which is close to what the calculation and scenario tools need to do. I have also helped construction companies work in an Agile way, so I am used to working with many different people: contractors, designers, logistics teams, recyclers and local governments. With my background in civil and environmental engineering and my GIS skills, I can help to build the tool and to make sure people will really use it.

For the last while I have taken a break from work to take care of my newborns. During this time I kept learning on my own about circular construction, reverse logistics and GIS. This EngD is the perfect way for me to come back to work. I can use what I already know, learn more, and this time apply it to a real project in industry and not only in theory.

I would really like to do this at the University of Twente. I like that the university works closely with industry, and that the EngD combines study with a real design project together with partners like Dura Vermeer and TNO. I learn best when I can use new methods straight away and improve them with the people I work with.

I would be very happy to work on CIRCOLOGIC and to grow as an engineer who builds tools that people actually use. I would be glad to talk more about how I can fit into the project. Thank you for reading my application.

Yours sincerely,

Pooja Awasare